

Note

Classification, types and patterns of fever :

- Fevers can be arbitrary classified into acute, sub-acute and chronic fevers based on duration.
- Acute fevers (<7 days in duration) are characteristics of infectious diseases such as malaria and viral-related upper respiratory tract infection
- Sub acute fevers (usually not more than 2 weeks in duration) may be seen in cases of typhoid fever and intra-abdominal abscess, among others
- Chronic or persistent fevers (>2 weeks duration) are typical of chronic bacterial infections such as tuberculosis, viral infections like HIV, cancers and connective tissue diseases
- However, any cause of acute fever can become persistent or chronic if untreated.
- Based on the height of body temperature, fever can also be classified into low grade, moderate grade, high grade and hyperpyrexia
- The height of body temperature may have some diagnostic and prognostic implications.
- Some studies have attributed high grade fevers in infants to serious bacterial infections, although others have also shown that children with high fevers are at equally high risk for serious bacterial infections and for viral illness.

Normal and febrile body temperature ranges (rectal temperatures).

Body temperature	$^{\circ}\text{C}$	$^{\circ}\text{F}$
Normal	37-38	98.6-100.4
Mild/low grade fever	38.1-39	100.5-102.2
Moderate grade fever	39.1-40	102.2-104.0
High grade fever	40.1-41.1	104.1-106.0
Hyperpyrexia	>41.1	>106.0

Three major fever types

Three major fever types have been described including sustained/continuous fever, intermittent fever and remittent fever

Continuous or sustained fever

- Continuous or sustained fever is defined as fever that does not fluctuate more than about 1°C (1.5°F) during 24 h, but at no time touches normal.
- Continuous fevers are characteristics of lobar and gram-negative pneumonia, typhoid, acute bacterial meningitis, urinary tract infection, among others
- Fever characterised by slow stepwise temperature rise and a high plateau are classical of typhoid fever.
- Fever associated with relative bradycardia (temperature pulse dissociation or Faget's sign) is a feature of untreated typhoid, leishmaniasis, brucellosis, Legionnaire's disease and psittacosis, Yellow Fever, among others.

Intermittent fever

- Intermittent fever is defined as fever present only for several hours during the day
- This pattern of fever can be seen in malaria, pyogenic infections, tuberculosis (TB), schistosomiasis, lymphomas, leptospira, borrelia, kala-azar, or septicemia

- Sources of continuous, intermittent or transient bacteraemia may lead to continuous, intermittent or transient fevers respectively.
- In malaria, depending on the species of parasite, fever can occur with a periodicity of 24 h (quotidian - due to *Plasmodium falciparum*), 48 h (tertian - *Plasmodium ovale* and *vivax*), or 72 h (quartan - *Plasmodium malariae*).
- The Pel-Epstein's fever is an intermittent low grade fever characterized by 3-10 days of fever with subsequent afebrile periods of 3-10 days.
- It is thought to be a typical but rare manifestation of Hodgkin's lymphoma.

Remittent fever

- Remittent fever is defined as fever with daily fluctuations exceeding 2°C but at no time touches normal.
- Remittent fevers are often associated with infectious diseases such as infective endocarditis, rickettsiae infections, brucellosis, among others.
- Relapsing fevers refer to those that are recurring and separated by periods with low-grade fever or no fever.
- Periodic or relapsing fevers are seen in malaria, lymphoma, borrelia, cyclic neutropenia, and rat-bite fever.
- Fever associated with night sweats has been described in infectious diseases such as TB, Nocardia, brucellosis, liver or lung abscess and sub-acute infective endocarditis, as well as in non-infectious diseases such as polyarteritis nodosa and cancers such as lymphomas.

Fever is then divided into three main categories

- Fever with localized signs
- Fever without localized signs
- Persistent fever of unknown cause (PUO)

Thermoregulation

- In healthy individuals, body temperature

varies in relation to several environmental and biological factors such as time of day, site of temperature measurements, level of physical activity, age, sex and race, among others.

- In spite of this variability, body temperature is tightly regulated within a fairly constant range - a thermal set point (recently renamed thermal balance point through the process of thermoregulation).
- The old view that temperature regulation functions as a unified control system with a single thermoregulatory centre has been rejected. Based on recent evidence, core temperature is regulated by various relatively independent thermoeffector loops, each having its own afferent and efferent branches.
- Hence, the regulation of body temperature is dependent on a thermoregulatory circuitry. However, the preoptic region of the anterior hypothalamus is still considered the major thermoregulatory centre in the CNS where peripheral and centrally generated temperature signals are received and integrated.
- The preoptic region consists of heat sensitive neurons, namely warm and cold sensitive neurons, which are activated or inhibited in response to temperature changes.
- In cold environments, decrease firing of warm sensitive neurons and increase firing of cold sensitive neurons leads to activation of heat gain mechanisms and prevention of heat loss (i.e. by skin vasoconstriction, piloerection, decrease sweating, increased muscle contraction, non-shivering thermogenesis, and seeking warm clothing or environments).
- In hot environments, increase firing of warm sensitive neurons and decrease firing of cold sensitive neurons leads to activation of heat loss mechanisms and inhibition of heat gain mechanisms (i.e. by sweating, removal of clothing or seeking cold environments).

- Ultimately, body temperature is kept within a normal regulated limit by a delicate balance between heat loss and heat gain.

Defining normal and febrile body temperatures

Fever: Temperature Measurement

- Rectal temperature measurement is considered to be the gold standard for children 3 years of age or younger. The most widely accepted definition of fever is rectal temperature of 38°C (100.4°F) or higher. It is important to consider that infants, especially those younger than 2 months of age, may have a blunted febrile (or hypothermic) response to infection. Hence, lack of fever should not be used as a criterion for ruling out infection in infants. Although rectal temperature measurement is the gold standard, it should be avoided in neutropenic immunocompromised patients, in whom rectal manipulation may seed the blood with bacteria.
- Oral thermometry can be considered for cooperative patients who are older than 4-5 years of age.
- Axillary temperatures are commonly done and tympanic membrane and temporal artery temperatures are newer modalities with some studies examining their reliability.
- Axillary temperatures are less precise than rectal temperatures. There is a correlation between axillary and rectal temperature measurements; the axillary temperature is usually $0.5\text{--}0.85^{\circ}\text{C}$ lower.
- Tympanic membrane thermometers are often inaccurate in children. Temporal artery temperature measurement correlates well with rectal temperature in some studies, but has been shown to be inferior when patients are febrile. It can be considered in settings when children are not likely to be febrile and are over 3 months of age. When detection of fever is critical for diagnosis and management, rectal temperatures should

be used in the child 3 years of age and younger.

- In the geriatric group (>65 years), who are likely to have lower body temperatures, IDSA defines fever as single oral temperature $>100^{\circ}\text{F}$ ($>37.8^{\circ}\text{C}$) or repeated oral temperatures $>99^{\circ}\text{F}$ ($>37.2^{\circ}\text{C}$) or rectal temperatures $>99.5^{\circ}\text{F}$ ($>37.5^{\circ}\text{C}$) or an increase in temperature of $>2^{\circ}\text{F}$ ($>1.1^{\circ}\text{C}$) over the baseline temperature.
- Fever has both positive and negative effects. High body temperatures may impair the reproduction and survival of some invading microorganisms by decreasing required nutrients, such as free iron, or by increasing immunologic responses such as phagocytosis.
- However, at extremely high temperatures, immunologic responses may be impaired. Fever increases the basal metabolic rate by 10-12% for each degree Celsius elevation of temperature. This increases oxygen consumption, carbon dioxide production, and fluid and caloric needs. Fluid requirements increase $100\text{ mL/m}^2\text{/day}$ for each 1°C rise in temperature above 37.8°C .
- Heat illness must be distinguished from fever as a cause for elevated body temperature.
- In heat illness, there is an unregulated rise in body temperature, despite the fact that the hypothalamic set point is normal. It can result from excessive heat production or inadequate heat dissipation.
- Temperatures may reach extreme heights and can result in multiorgan dysfunction and death. Hence restoration of normal body temperature in heat illness is mandatory.

Physical Examination

- Many children will have a source for fever identified on their history and/or physical examination.
- If no focus of infection is found on the physical examination, the clinician must rely

on history and observation to determine the appropriate next steps. The child may appear ill or well. Ill-appearing children are typically lethargic or irritable. They may show signs of shock, including weak peripheral pulses, tachycardia, poor perfusion, respiratory distress, mottling, cyanosis, or decreased mental status. After thorough clinical and laboratory evaluation, ill-appearing children should be admitted to the hospital, and will likely need empiric antibiotic treatment.

Causes of Hyperthermia

Excessive Heat Production

- Exertion
- Heat stroke (exertion)
- Malignant hyperthermia (anesthesia induced)
- Neuroleptic malignant syndrome
- Catatonia
- Tetanus
- Status epilepticus
- Delirium
- Endocrine disorders (hyperthyroidism, pheochromocytoma)
- Drugs (cocaine, amphetamines, ephedrine, phencyclidine, tricyclic antidepressants, lysergic acid diethylamide (LSD), lithium, thyroid hormone, salicylates)

Diminished Heat Dissipation

- Heat stroke
- Occlusive dressings
- Dehydration
- Extensive burns (including severe sunburn)
- Anhidrotic ectodermal dysplasias
- Anticholinergic-like drugs (atropine, antihistamines, phenothiazines, tricyclic antidepressants)
- Autonomic neuropathy
- Spinal cord level paralysis (spinal crisis)
- Possible overbundling (especially in a warm environment)
- Therapeutic hyperthermia

Hypothalamic Dysfunction (Usually associated with hypothermia)

- Stroke
- Encephalitis
- Granulomatous processes (sarcoid, tuberculosis, eosinophilic)
- Trauma
- Central: idiopathic
- Phenothiazines
- Hemorrhage

Etiology and pathogenesis of Jvara

मिथ्याहारविहारभ्यां दोषा ह्यामाशयाश्रयाः ।
बहिर्निस्स्य कोष्ठाग्निं ज्वरदाः स्यू रसानुगाः ॥

The *Doṣas* aggravated in the *Āmāśaya* by the consumption of contrary foods and activities leading to the expulsion of *Agni* from the *Āmāśaya* associated with *rasa dhātu* leading to development of fever.

Note-

Causes of fever : Fever is the result of your immune system's response to a foreign invader. These foreign invaders include viruses, bacteria, fungi, drugs, or other toxins.

Fever : the role of pyrogens and cryogens

- The initiation, manifestations and regulation of the febrile response are dependent on the pyrogenic and anti-pyretic properties of various exogenous and endogenous substances.
- While pyrogens directly or indirectly lead to fever, cryogens prevent excessive temperature elevation. It is the balance in the interactions between pyrogens and cryogens that determine the height and duration of the febrile response to any immune challenge.

Pyrogens

- The foreign invaders are considered fever-producing substances (called pyrogens), which trigger the body's immune response. Pyrogens signal the hypothalamus in the brain to increase the body temperature set point in order to help the body fight off the infection.

- Substances that cause fever are called pyrogens.
- Pyrogens can come from inside or outside the body.
- Microorganisms and the substances they produce (such as toxins) are examples of pyrogens formed outside the body.
- Pyrogens are classified into exogenous (produced outside the host) and endogenous (produced within the host) pyrogens based on their site of production.
- Pyrogens formed inside the body are usually produced by monocytes and macrophages (two types of white blood cells). Endogenous pyrogens are mainly pyrogenic cytokines including interleukins (IL) 6, IL-1, interferon gamma (INF- γ) and ciliary neurotropic factor (CNTF) and tumour necrosis factor (TNF α), among others.
- Pyrogens from outside the body can cause fever by stimulating the body to release its own pyrogens or by directly affecting the area of the brain that controls temperature. Exogenous pyrogens are, essentially, part or whole micro-organisms or products of micro-organisms such as toxins.
- Infection is not the only cause of fever.
- Fever may also result from inflammation, a reaction to a drug, an allergic reaction, autoimmune disorders (when the body produces abnormal antibodies that attack its own tissues), and undetected cancer (especially leukemia, lymphoma, or kidney cancer).

Cryogens

- Cryogens include anti-inflammatory cytokines (e.g. IL-10), hormones (e.g. α -melanocyte stimulating hormone, corticotrophin and corticotrophin releasing hormone) and many other neuroendocrine products (e.g. neuropeptide Y, bombesin, and thyroliberin), cytochrome P-450 (P-450), among others.
- They exert their anti-pyretic effects by inhibiting synthesis of pyrogenic cytokines (e.g. glucocorticoids), cytokine receptors

blockade (e.g. IL-1 receptor antagonist), and increasing heat loss by enhancing sensitivity of warm sensitive neurons (e.g. bombesin), among other mechanisms. These endogenous antipyretic systems protect the host against the destructive consequences of unchecked fever.

Many disorders can cause fever. They are broadly categorized as :

- Infectious (most common)
- Neoplastic (cancer)
- Inflammatory
- An infectious cause is highly likely in adults with a fever that lasts 4 days or less.
- A noninfectious cause is more likely to cause fever that lasts a long time or returns.

Many cancers cause fever

- Inflammatory disorders that cause fever include joint, connective tissue, and blood vessel disorders such as rheumatoid arthritis, systemic lupus erythematosus (lupus), and giant cell arteritis.
- Also, an isolated, short-lived (acute) fever in people with cancer or a known inflammatory disorder is most likely to have an infectious cause.
- In healthy people, an acute fever is unlikely to be the first sign of a chronic illness.
- Drugs sometimes cause fever. For example, beta-lactam antibiotics (such as penicillin) and sulfa drugs can trigger a fever.
- Drugs that can cause an extremely high temperature include certain illicit drugs (such as cocaine, amphetamines, or phencyclidine), anesthetics, and antipsychotic drugs.

Most common causes

- Virtually all infectious disorders can cause fever.
- But overall, the most likely infectious causes are
- Upper and lower respiratory tract infections
- Gastrointestinal infections
- Urinary tract infections (UTIs)

- Skin infections
- Most acute respiratory tract and gastrointestinal infections are viral.

Risk factors

- Certain conditions (risk factors) make people more likely to have a fever. These factors include the following:

- The person's health status
- The person's age
- Certain occupations
- Use of certain medical procedures and drugs
- Exposure to infections (for example, through travel or contact with infected people, animals, or insects)

Some Causes of Fever Based on Risk Factors

Risk Factor	Cause
None (healthy)	Upper or lower respiratory tract infection Gastrointestinal infection Urinary tract infection Skin infection
Hospitalization	Infection related to a catheter inserted in a vein (IV catheter infection) Urinary tract infection, particularly in people with a urinary catheter Pneumonia, particularly in people on a ventilator Atelectasis (collapse of part of a lung due to an airway blockage, rather than an infection) Infection or a pocket of blood (hematoma) at the site of surgery Deep vein thrombosis or pulmonary embolism Diarrhea (due to <i>Clostridioides difficile</i> induced colitis) Drugs Transfusion reaction Pressure sores
Travel to areas where an infection is common (endemic areas)	Malaria Viral hepatitis Disorders that cause diarrhea Typhoid fever Zika virus, chikungunya, Ebola virus, Japanese encephalitis, rabies, measles, and yellow fever Dengue fever Fungal infections such as coccidioidomycosis, blastomycosis, or histoplasmosis Rickettsial infections (such as African tick-bite fever and Mediterranean spotted fever)
Exposure to insects or animals that carry disease-causing organisms, called vectors (in the United States)	Ticks: Rickettsiosis, ehrlichiosis or anaplasmosis, Lyme disease, babesiosis, relapsing fever, or tularemia Mosquitoes: Encephalitis caused by arboviruses, malaria, or Zika virus infection Wild animals: Tularemia, rabies, or hantavirus infection Fleas: Plague Domestic animals: Brucellosis, cat-scratch disease, Q fever, or toxoplasmosis Birds: Psittacosis Reptiles: Salmonella infection Bats: Rabies or histoplasmosis

A weakened immune system (immunocompromise)	Viruses: Varicella-zoster virus infection or cytomegalovirus infection Bacteria: Infection due to pneumococci, meningococci, <i>Staphylococcus aureus</i>, <i>Pseudomonas aeruginosa</i>, <i>Nocardia</i>, or mycobacteria Fungi: Infection due to <i>Candida</i>, <i>Aspergillus</i>, <i>Histoplasma</i>, <i>Coccidioides</i>, microsporidia, <i>Pneumocystis jirovecii</i>, or fungi that cause mucormycosis Parasites: Infection due to <i>Toxoplasma gondii</i>, <i>Strongyloides stercoralis</i>, <i>Cryptosporidium</i>, or <i>Cystoisospora belli</i>
Drugs that can increase heat production	Amphetamines Cocaine Phencyclidine (PCP) Methylenedioxymethamphetamine (MDMA, or Ecstasy) Antipsychotic drugs Anesthetics
Drugs that can trigger fever	Beta-lactam antibiotics (such as penicillin) Sulfa drugs Phenytoin Carbamazepine Procainamide Quinidine Amphotericin B Interferons (drugs that are based on substances produced by the immune system and that help block the reproduction of viruses)

Fever of unknown origin (FUO)

A fever of unknown origin may be diagnosed when :

- People have a fever of at least 101 °F (38.3 °C) for several weeks
- Extensive investigation does not detect a cause

Pathophysiology of the febrile response

- The development of the febrile response is akin to the normal thermoregulatory processes that follow exposure to cold temperatures. However, in fever the thermal balance point is reset to a higher level such that normal peripheral and central body temperatures are now sensed as cold temperature signals by the thermoregulatory circuitry.
- Consequently, fever is different from heat stroke and hyperthermia where body temperature is elevated without a

corresponding elevation of the thermal balance point.

- Like thermoregulation, evolving evidence suggest that the generation of fever follows multiple independent afferent and efferent mechanisms depending on the site, nature and severity of inflammation.
- The various biological molecules involved in the generation of the febrile response and the pathways implicated in these responses are discussed in the following section.

The fever pathways

Fever signals carried by exogenous and endogenous pyrogens ultimately lead to reset of the thermoregulatory circuitry via two basic pathways, namely the humoral and neural pathways.

The humoral pathway

- In this pathway, fever signals are carried by components of microbial products named

pathogen associated molecular patterns (PAMPS) or by pyrogenic cytokines.

- The febrile response is characterised by an early rapid phase and a delayed late phase
- The second humoral pathway is directed by circulating pyrogenic cytokines. They transmit fever signals to the thermoregulatory circuitry by both indirect and direct pathways.

The neural pathway

Peripheral fever signals can communicate with the CNS through peripheral nerves such as cutaneous sensory nerves and the vagus nerve. The activation of the neural pathway is believed to be another mechanism by which fever is rapidly initiated.

Mechanisms underlying fever patterns

- The mechanisms underlying the specificity of the fever patterns to certain diseases are not completely understood. For some infectious diseases it may be related to the life cycle of causative agent. For instance, in malaria, parasites are released into the circulation after 48-72 h of erythrocytic cycle of *Plasmodium falciparum*/ovale/vivax.
- Released parasites activate pyrogenic cytokines which then lead to cycles of fever every 48-72 h (tertian fever).
- However, *Plasmodium falciparum* in contrast to other species may infect multiple red cells in a non-selective manner with each having independent erythrocytic parasite life cycles.
- Consequently, fever due to this parasite is often quotidian (daily fever spikes).
- *Plasmodium vivax*/ovale and *P. malariae* infects young and senescent erythrocytes that rupture to release merozoites (pyrogens) 72- and 96-h respectively. These events partly account for the cyclical nature of fever in these malarial fevers.
- A down regulation of cytokine release after repeated exposure to pyrogens such as LPS may lead to fever remittance or intermittence

- Recurrent fevers may be due to partial treatment of deep seated infections like abscesses or due to recurrent exposure to new antigens (e.g. allergens in hypersensitivity pneumonitis).
- Such drug (allergens) may manifest with eosinophilia in patients with drug fever. In cyclic neutropenic fevers, febrile episodes correspond to periods of neutropenia and are due to repeated bacterial infections.
- In cancers and pulmonary embolism, fever recurrence has been attributed, partly, to tissue necrosis as phagocytosis of necrotic tissue leads to intermittent release of pyrogenic cytokines.
- Recurrent fevers may be related to the pathogenesis of the disease as exemplified in relapsing fevers due to spirochetes where episodic spirochetemia lead to episodes of fever accompanied by fever free periods which coincides with disappearance of spirochetes from the circulation.
- Night sweats are common in healthy adults but they become clinically significant if associated with fever and drenching.
- They may be attributed to pyrogenic properties of certain diseases which result in early morning fever spikes accompanied hours later by fever remission presenting as night sweats.

Significance of fever patterns

- In clinical practice, many typical fever patterns described above are rarely observed.
- This is as a result of several confounding factors such as ingestion of antipyretics, partial treatment infectious disease-related fevers with antimicrobials and failure to mount fever in the face active infection (for instance the elderly, immunocompromised and severely malnourished individuals may not mount fever).
- Furthermore, the occurrence of concurrent infections by multiple agents in an individual

may alter typical fever patterns associated with specific infectious diseases. For these reasons, the utility of fever patterns in clinical diagnosis is often limited.

- However, when observed, typical fever patterns may provide some useful diagnostic clues for infectious diseases without localising signs such as malaria and typhoid. They may also prove useful in eliminating unlikely diagnosis while directing possible diagnostic investigations to be undertaken for confirmation. In poor resource settings, analysis of fever patterns may provide diagnostic clues for difficult to diagnose infectious diseases especially when appropriate diagnostic investigations are unavailable or unhelpful.
- In the evaluation of pyrexia of unknown origin (PUO) for instance, the presence of low grade intermittent fever and night sweats without localising sign is a useful clue to investigate for extrapulmonary tuberculosis or to initiate empirical anti-tuberculosis therapy in tuberculosis endemic regions.
- The nature of fever defervescence may also provide some diagnostic clues. For instance, in differentiating malaria from typhoid fever, effective anti-malarial therapy leads to fever defervescence by crisis (within hours) whereas in typhoid fevers resolution occurs by lysis (gradually over days) following effective antibiotics.
- However, failure of fever to resolve by crisis does not exclude pre-existing malaria as malaria and typhoid fever may occur concurrently, especially in the tropics.
- It has been suggested that fevers due to cancers can be distinguished from those due to infectious diseases using the Naprosyn test.
- When Naprosyn (Naproxen 375 mg twice daily), or other NSAIDS, is given for 3 days, fevers due to cancers display a rapid and sustained decline while little or no change is observed in fevers due to infectious diseases.

- However, Naprosyn test is unhelpful in differentiating neoplastic from noninfectious disorders such as connective tissue diseases.
- Similarly, elevated serum procalcitonin level has been used to differentiate infectious from non-infectious causes of fever with variable results.

Diseases called 'fever'

- Based on its fundamental role in explaining the manifestations of diseases as well as its prominence as a feature of many diseases of man and animals, fever is often included in the appellation of many diseases.
- The naming of diseases called fever are usually based on specific epidemiological and clinical characteristics of the disease including: the way the disease is acquired (e.g. Hay fever is a febrile illness due to exposure to 'hay' in predisposed individuals); the transmitting vector/agent (e.g. cat scratch fever is an infectious disease due to a bite or 'scratch from a cat'); the prominent clinical manifestation (e.g. yellow fever is a viral illness so named because jaundice or 'yellow eyes' is a prominent manifestation in severe cases); the geographical location where the disease was first discovered (e.g. Lassa fever is a haemorrhagic viral infection so named because it was first described in 1969 in a village named Lassa, in Borno state, Nigeria; where it is highly prevalent (e.g. Familial Mediterranean fever is a hereditary febrile disease common among people in the Mediterranean region); where a major outbreak occurred (e.g. Pontiac fever, a non-pneumonic variant of Legionella infection is named after an outbreak in Pontiac, Michigan, in 1968).

Measles

Before immunisation campaigns, measles occurred in almost 100% of children world wide. The WHO has set the objective of eradicating measles globally by 2010, using the live attenuated

vaccine. However, vaccination of only 70-80% of the population, as is currently the case in the UK, for examples, is insufficient to prevent outbreaks in older children and adults, who are more susceptible to complications. Natural illness produces lifelong immunity.

Clinical features

Incubation period-6-19 days

It spreads by respiratory droplets.

Prodromal symptoms

1. Upper respiratory symptoms
2. Conjunctivitis
3. Presence of Koplik's spots on the internal buccal mucosa.

General signs & symptoms

1. The small white spots surrounded by erythema are pathognomonic of measles.
2. As natural antibody develops the maculopapular rash appears, spreading from the face to the extremities.
3. General Lymphadenopathy
4. Diarrhoea

Complications

1. Otitis media
2. Bacterial pneumonia
3. Clinical encephalitis in 0.1% of children
4. Subacute sclerosing panencephalitis (SSPE)

Investigations & Diagnosis

1. Diagnosis is by clinical features
2. Detection of antibody-Serum IgM, Sero Conversion or Salivary IgM.

Note :

1. Disease is more severe and prolonged in adults and complications include pneumonitis, hepatitis and encephalitis.
2. Measles is a serious disease in the malnourished, vitamin deficient or immunocompromised, in whom the typical rash may be missing and persisting infection with a giant cell pneumonitis or encephalitis may occur.
3. Death usually results from bacterial

superinfection such as pneumonia, diarrhoea or noma/cancrum oris.

Chickenpox (varicella)

1. Varicella zoster virus (VZV) is a dermatropic and neurotropic virus that produces primary infection, usually in childhood, which may reactivate in later life.
2. VZV is spread by aerosol and direct contact.
3. It is highly infectious to non-immune individuals.
4. Disease in children is usually well tolerated.
5. Manifestations are more severe in adults, pregnant women and the immunocompromised.

Clinical features

1. The incubation period is 11-20 days
2. Vesicular eruption begins, often on mucosal surfaces first, followed by rapid dissemination in a centripetal distribution (most dense on trunk and sparse on limbs).
3. New lesions occur every 2-4 days and each crop is associated with fever.
4. The rash progresses from small pink macules to vesicles and pustules within 24 hours.
5. Infectivity lasts from up to 4 days (but usually 48 hours) before the lesions appear until the last vesicles crust over.

Complications

6. Due to intense itching, secondary bacterial infection from scratching is the most common complication of primary chickenpox.
7. Self-limiting cerebellar ataxia and encephalitis are rare complications.

Risk Factors

8. Adults, pregnant women and the immunocompromised are at increased risk of visceral involvement, which presents as pneumonitis, hepatitis or encephalitis.
9. Pneumonitis can be fatal and is more likely to occur in smokers.
10. Maternal infection in early pregnancy carries a 3% risk of neonatal damage with

developmental abnormalities of eyes, CNS and limbs.

11. Chickenpox within 5 days of delivery leads to severe neonatal varicella with visceral involvement and haemorrhage.

Diagnosis

1. Diagnosis is primarily clinical, by recognition of the rash.
2. If necessary, this can be confirmed by detection of antigen (direct immunofluorescence), DNA (PCR) or by viral culture of aspirated vesicular fluid.
3. Serology is used to identify seronegative individuals at risk of infection.

Herpes Simplex Virus Infections

Etiologic Agent

- Herpes simplex viruses (HSV-1, HSV-2; *Herpesvirus hominis*) produce a variety of infections involving mucocutaneous surfaces, the central nervous system (CNS), and—on occasion—visceral organs.
- Prompt recognition and treatment reduce the morbidity and mortality rates associated with HSV infections.

Pathogenesis

Exposure to HSV at mucosal surfaces or abraded skin sites



Permits entry of the virus into cells of the epidermis and dermis



Initiation of viral replication starts



During the initial phase of infection, viral replication occurs in ganglia and contiguous neural tissue.



Virus then spreads to other mucocutaneous surfaces through centrifugal migration of infectious virions via peripheral sensory nerves.



This mode of spread helps explain the large surface area involved, the high frequency of new lesions distant from the initial crop of vesicles that is characteristic in patients with primary genital or oral-labial HSV infection,



And the ability to recover virus from neural tissue distant from neurons innervating the inoculation site.



Contiguous spread of locally inoculated virus also may take place and allow further mucosal extension of disease.

Clinical Spectrum

1. HSV has been isolated from nearly all visceral and mucocutaneous sites.
2. The incubation period ranges from 1 to 26 days (median, 6–8 days).
3. Both viral subtypes can cause genital and oral-facial infections, and the infections caused by the two subtypes are clinically indistinguishable.
4. Genital HSV-2 infection is twice as likely to reactivate and recurs 8–10 times more frequently than genital HSV-1 infection.
5. Oral-labial HSV-1 infection recurs more frequently than oral-labial HSV-2 infection.

Oral-Facial Infections

- Gingivostomatitis and pharyngitis are the most common clinical manifestations of first-episode HSV-1 infection, while recurrent herpes labialis is the most common clinical manifestation of reactivation HSV-1 infection.
- HSV-1 and varicella-zoster virus (VZV) have been implicated in the etiology of Bell's palsy (flaccid paralysis of the mandibular portion of the facial nerve).

Genital Infections

1. First-episode primary genital herpes is characterized by fever, headache, malaise, and myalgias.
2. Pain, itching, dysuria, vaginal and urethral discharge, and tender inguinal

lymphadenopathy are the predominant local symptoms.

Herpetic Whitlow

Herpetic whitlow—HSV infection of the finger—may occur as a complication of primary oral or genital herpes by inoculation of virus through a break in the epidermal surface or by direct introduction of virus into the hand through occupational or some other type of exposure.

Herpes Gladiatorum

HSV may infect almost any area of skin. Mucocutaneous HSV infections of the thorax, ears, face, and hands have been described among wrestlers.

Eye Infections

HSV infection of the eye is the most common cause of corneal blindness in the United States. HSV keratitis presents as an acute onset of pain, blurred vision, chemosis, conjunctivitis, and characteristic dendritic lesions of the cornea.

Central and Peripheral Nervous System Infections

HSV accounts for 10–20% of all cases of sporadic viral encephalitis.

Visceral Infections

HSV infection of visceral organs usually results from viremia, and multiple-organ involvement is common. Occasionally, however, the clinical manifestations of HSV infection involve only the esophagus, lung, or liver.

Neonatal HSV Infections

Neonates (infants younger than 6 weeks) have the highest frequency of visceral and/or CNS infection.

HSV in Pregnancy

The risk of mother-to-child transmission of HSV in the perinatal period is highest when the infection is acquired near the time of labor—that is, in previously HSV-seronegative women.

Diagnosis-Investigations

1. Both clinical and laboratory criteria are useful

for diagnosing HSV infections.

2. Staining of scrapings from the base of the lesions with Wright's, Giemsa's (Tzanck preparation), or Papanicolaou's stain to detect giant cells or intranuclear inclusions of *Herpes virus* infection
3. HSV infection is best confirmed in the laboratory by detection of virus, viral antigen, or viral DNA in scrapings from lesions.
4. HSV DNA detection by PCR is the most sensitive laboratory technique for detecting mucosal or visceral HSV infections
5. HSV causes a discernible cytopathic effect in a variety of cell culture systems, and this effect can be identified within 48–96 h after inoculation.
6. Spin-amplified culture with subsequent staining for HSV antigen has shortened the time needed to identify HSV to <24 h.

Laboratory confirmation permits subtyping of the virus; information on subtype may be useful epidemiologically and may help to predict the frequency of reactivation after first-episode oral-labial or genital HSV infection.

Herpes zoster

Herpes zoster is usually a disease of older people. It is commonly known as shingles and also known as zona, is a viral disease characterized by a painful skin rash with blisters in a limited area on one side of the body, often in a stripe. Throughout the world, the incidence rate of herpes zoster every year ranges from 1.2 to 3.4 cases per 1,000 healthy individuals, increasing to 3.9–11.8 per year per 1,000 individuals among those older than 65 years. It is a self-limiting disorder but unlike chicken pox the rash is very painful and even after it disappears, the pain may persist specially in the elderly due to post herpetic neuralgia. .

Signs and Symptoms

- Headache
- fever
- malaise
- burning pain

- itching
- hyperesthesia or paresthesia

Herpes zoster may have additional symptoms, depending on the dermatome involved. *Herpes zoster ophthalmicus* involves the orbit of the eye and occurs in approximately 10 to 25% of cases. It is caused by the virus reactivating in the ophthalmic division of the trigeminal nerve. In a few patients, symptoms may include conjunctivitis, keratitis, uveitis, and optic nerve palsies that can sometimes cause chronic ocular inflammation, loss of vision, and debilitating pain. *Herpes zoster oticus*, also known as Ramsay Hunt syndrome type II, involves the ear. It is thought to result from the virus spreading from the facial nerve to the vestibulo-cochlear nerve. Symptoms include hearing loss and vertigo (rotational dizziness).

Coxsackieviruses

- Coxsackieviruses belong to the family Picornaviridae and the genus Enterovirus, which also includes poliovirus and echovirus.
- Enteroviruses are among the most common and important human pathogens with 9 different species classified (Enterovirus A-H and I).

Pathophysiology

- Coxsackieviruses primarily are transmitted via the fecal-oral route and respiratory aerosols, although transmission via fomites also is possible.
- Viral replication initially occurs in the upper respiratory tract, particularly the tonsils, and the distal small bowel.
- Viremia with systemic dissemination then occurs and involves the reticuloendothelial system, which leads to further replication in other anatomic locations and organs and ultimately to development of symptoms.
- Central nervous system (CNS) invasion is proposed to occur from viral migration along peripheral and central nerves into the CNS.
- Coxsackieviruses have been identified in the respiratory tract up to 3 weeks after initial

infection and in feces up to 8 weeks after initial infection. Innate, humoral and cell-mediated immunity all play a role in the body's response to enteroviral infection. However, the infection frequently is eradicated before antibody production occurs.

Clinical Presentation

- More than 90% of coxsackieviruses infections are asymptomatic or cause nonspecific febrile illnesses.
- In neonates, they are the most common cause of febrile illnesses during the summer and fall months.
- As measured year round, thirteen percent of newborns with fever in the first month of life were noted to have an enteroviral infection.

In addition to nonspecific febrile illnesses, various well-described syndromes also have been associated with coxsackievirus infections.

- Aseptic meningitis
- Encephalitis
- Other neurologic syndromes
- Rarely, coxsackieviruses have been implicated in other sporadic neurologic syndromes such as acute flaccid myelitis (AFM) that closely mimics poliovirus infection.
- Myopericarditis
- Diabetes
- Exanthems and enanthems
- Two of the most distinctive rash syndromes caused by coxsackieviruses are hand, foot, and mouth disease (HFMD) and herpangina.
- Epidemic pleurodynia
- Acute hemorrhagic conjunctivitis
- Sepsis

Ancillary investigations

- Enteroviruses can be excreted in human feces for up to 3 months after infection.
- However, a clinically identifiable syndrome correlates with the acute phase of infection, during which time virus can be found in the throat, blood, and various organs.

- There are no confirmatory laboratory tests, procedures, or imaging that are used in routine clinical practice for HFMD or herpangina.
- Diagnosis for these conditions mainly is based on clinical presentation and assessment.

Rubella (German measles)

- Rubella is an endemic exanthem in countries without universal vaccination.
- In non-immunised communities 80-85% of young adults have evidence of past infection.

Aetiology

- Rubella and congenital rubella syndrome are caused by rubella virus.
- Only one antigenic type of rubella virus is available
- The humans are the only natural hosts

Clinical features

1. Rubella is spread by respiratory droplet, with infectivity from up to 10 days before to 2 weeks after the onset of the rash.
2. The incubation period is 15-20 days.
3. In childhood, most cases are subclinical, although clinical features may include fever, maculopapular rash spreading from the face, and lymphadenopathy.
4. Complications are rare but include thrombocytopenia and hepatitis.
5. Encephalitis and haemorrhage are occasionally reported.
6. In adults, arthritis involving hands or knees is relatively common, especially in women.

Rubella infection: risk of congenital malformation

Stage of gestation Likelihood of malformations

1-2 months	65-85% chance of illness, multiple defects/ spontaneous abortion
3 months	30-35% chance of illness, usually a single congenital defect (most frequently deafness, cataract, glaucoma, mental retardation, congenital heart disease, especially

pulmonary stenosis, patent ducts arteriosus)

4 months	10% risk of congenital defects, most commonly deafness
> 20 weeks	Occasional deafness

Diagnosis

1. Laboratory confirmation of rubella is required if there has been contact with a pregnant woman.
2. This is achieved either by detection of rubella IgM in serum or by IgG seroconversion.
3. In the exposed pregnant woman, absence of rubella-specific IgG confirms the potential for congenital infection.

Hand, foot, and mouth disease (HFMD)

- HFMD often affects children and easily spreads to close contacts.
- Patients present with a sore throat and painful lesions in the mouth.
- Vesicles that coalesce, form bullae, and then ulcerate, occur on the buccal mucosa and tongue. Seventy-five percent of patients have simultaneous peripheral cutaneous lesions.
- HFMD also can present more atypically, including as eczema coxsackium, with erosions or bullous lesions.
- Biopsy reveals intracytoplasmic viral particles. Coxsackieviruses A16, A6, and A10 are among the most common types implicated in HFMD.
- Numerous cases of a more severe HFMD caused by coxsackievirus A6 were reported between 2004 and 2011 in several Asian and European countries. Additionally, between 2011 and 2012 in the United States, among cases of severe HFMD that were reported, 74% had a positive PCR test for coxsackievirus A6; about 25% of reported cases occurred in adults.

Filariasis

Filariasis is a group of disorders caused by infection with the thread-like nematodes. Lymphatic filariasis is caused by *Wuchereria bancrofti*

and *Brugia malayi*. Light infections may be completely asymptomatic.

Symptoms

The clinical manifestations are:

- o Low grade fever which may be accompanied by chills and sweats
- o lymphangitis, involving the legs more frequently than the arms
- o lymphadenitis, involving inguinal, femoral and epitrochlear nodes
- o dilatation of lymphatic which may rupture
- o chyluria
- o lymphedema of one or more extremities or other parts
- o episodes of secondary bacterial infection
- o tropical eosinophilia.

Diagnosis: Microfilariae may be detected in peripheral smear (night sample), Serological tests are not reliable.

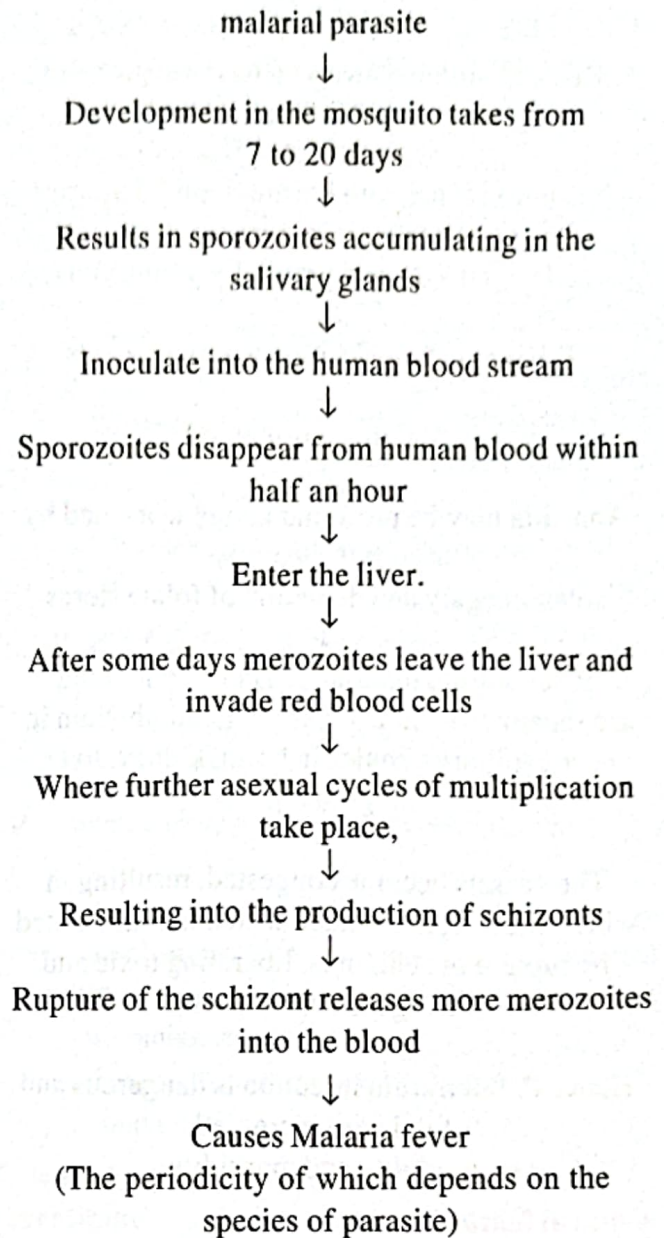
Malaria

1. Malaria is a protozoan disease transmitted by the bite of infected *Anopheles* mosquitoes
2. Malaria in humans is caused by plasmodium falcifarum, P.Vivax, P. Ovale, P. malariae and the predominantly simian parasite, P Knowles.
3. It is transmitted by the bite of female anopheline mosquitoes and occurs throughout the tropics and subtropics at altitudes below 1500 metres.
4. Travellers are susceptible to malaria.
5. A few people living near airports in Europe have acquired malaria from accidentally imported mosquitoes.

Pathogenesis

Life cycle of the malarial parasite

The female *Anopheles* mosquito becomes infected when it feeds on human blood containing gametocytes, the sexual forms of the



High yield information

P. Vivax, P. Ovale may persist in liver cells as dormant forms, hypnozoites, capable of developing into merozoites months or years later. Thus the first attack of clinical malaria may occur long after, the patient has left the endemic area, and the disease may relapse after treatment if drugs that kill only the erythrocytic stage of the parasite are given.

Plasmodium falcifarum and P. malariae have no persistent exoerythrocytic phase but recrudescence of fever may result from multiplication of parasites in red cells which have not been eliminated by treatment and immune processes.

Pathology

Red cells infected with malaria are prone to haemolysis



It is most severe with *Plasmodium falciparum* because it invades red cells of all ages but especially young cells



P. Vivax, *P. ovale* invade reticulocytes



P. Malariae normoblasts



Anaemia may be profound and is worsened by dyserythropoiesis, splenomegaly and depletion of folate stores



P. falciparum malaria, red cells containing trophozoites adhere to vascular endothelium in post-capillary venules in brain, kidney, liver, lungs and gut.



The vessels become congested, resulting in widespread organ damage which is exacerbated by rupture of schizonts, liberating toxic and antigenic substances



Hence *P. falciparum* infection is dangerous and fatal with increased mortality and morbidity

Clinical features***P. falciparum* infection**

1. This is the most dangerous of the malarias and patients are either 'killed or cured'
2. The onset is often insidious, with malaise; headache and vomiting associated with Cough and mild diarrhoea are also common.
3. The fever has no particular pattern.
4. Jaundice is common due to haemolysis and hepatic dysfunction.
5. The liver and spleen enlarge and may become tender.
6. Anaemia develops rapidly, as does thrombocytopenia
7. May rapidly develop dangerous complications

8. Cerebral malaria is manifested by confusion, seizures or coma, usually without localising signs.

9. Children die rapidly without any special symptoms other than fever. F

10. Immunity is impaired in pregnancy and the parasite can preferentially bind to a placental protein known as chondroilin sulphate A.

11. Abortion and intrauterine growth retardation from parasitisation of the maternal side of the placenta are frequent.

12. Previous splenectomy increases the risk of severe malaria.

***P. vivax* and *P. ovale* infection**

1. In many cases the illness starts with several days of continued fever before the development of classical bouts of fever on alternate days.
2. Fever starts with a rigor. The patient feels cold and the temperature rises to about 40°C.
3. After half an hour to an hour the hot or flush Phase begins.
4. It lasts several hours and gives way to profuse perspiration and a gradual fall in temperature.
5. The cycle is repeated 48 hours later.
6. Gradually the spleen and liver enlarge and may become tender.
7. Anaemia develops slowly.
8. Relapses are frequent in the first 2 years after leaving the malarious area and infection may be acquired from blood transfusion.

***P. malariae* infection**

1. This is usually associated with mild symptoms and bouts of fever every third day
2. Parasitemia may persist for many years with the occasional recrudescence of Fever or without producing any symptoms.
3. Chronic *P. malariae* infection causes glomerulonephritis and longterm nephrotic syndrome in children.

Investigations**Peripheral blood examination**

- Giemsa-stained thick and thin blood films

should be examined whenever malaria is suspected.

- In the thick film erythrocytes are lysed, releasing all blood stages of the Parasite.
- A thin film is essential to confirm the diagnosis, to identify the species of parasite and in *P. falciparum* infections.

Immunochromatographic tests

- Immunochromatographic tests for malaria antigens, such as Optimal (which detects the *Plasmodium* lactate dehydrogenase of several species)
- Parasight F (which detects the *P. falciparum*)

PfHRP2 dipstick or card test

- A drop of blood is placed on the stick or card, which is then immersed in washing solutions.
- Monoclonal antibody captures the parasite antigen and reads out as a colored band.

Disadvantages

- Detects only *Plasmodium falciparum*; remains positive for weeks after infection; does not quantitate *P. falciparum* parasitemia

Plasmodium LDH dipstick or card test

- A drop of blood is placed on the stick or card, which is then immersed in washing solutions.
- Monoclonal antibodies capture the parasite antigens and read out as colored bands. One band is genus specific (all malarias), and the other is specific for *P. falciparum*.

Disadvantages

Slightly more difficult preparation than PfHRP2 tests; may miss low-level parasitemia with *P. vivax*, *P. ovale*, and *P. malariae* and does not speciate these organisms; does not quantitate *P. falciparum* parasitemia

Microtube concentration methods with acridine orange staining

- Blood is collected in a specialized tube containing acridine orange, anticoagulant, and a float.

- After centrifugation, which concentrates the parasitized cells around the float, fluorescence microscopy is performed.

Full blood count and general blood picture

- Normochromic, normocytic anemia is usual.
- The leukocyte count is generally normal, although it may be raised in very severe infections.
- There is slight monocytosis, lymphopenia, and eosinopenia, with reactive lymphocytosis and eosinophilia in the weeks after the acute infection.
- The platelet count is usually reduced to $<10^5/\mu\text{L}$.

ESR and CRP

- The erythrocyte sedimentation rate, plasma viscosity, and levels of C-reactive protein and other acute-phase proteins are high.

Prothrombin and partial thromboplastin times

- Severe infections may be accompanied by prolonged prothrombin and partial thromboplastin times and by more severe thrombocytopenia.
- Levels of antithrombin III are reduced even in mild infection.

Electrolytes, blood urea nitrogen (BUN), and creatinine

- In uncomplicated malaria, plasma concentrations of electrolytes, blood urea nitrogen (BUN), and creatinine are usually normal.
- metabolic acidosis, with low plasma concentrations of glucose, sodium, bicarbonate, calcium, phosphate, and albumin together with elevations in lactate, BUN, creatinine, urate, muscle and liver enzymes, and conjugated and unconjugated bilirubin may be observed in severe malaria

Hypergammaglobulinemia

- Hypergammaglobulinemia is usual in immune and semi-immune subjects.

Urinalysis

- Urinalysis generally gives normal results.

Cerebrospinal fluid (CSF)

In adults and children with cerebral malaria, the mean opening pressure at lumbar puncture is 160 mm of cerebrospinal fluid (CSF); usually the CSF is normal or has a slightly elevated total protein level [<1.0 g/L (<100 mg/dL)] and cell count (<20 /L).

Influenza

Definition

- Influenza is an acute respiratory illness caused by infection with influenza viruses.
- The illness affects the upper and/or lower respiratory tract and is often accompanied by systemic signs and symptoms such as fever, headache, myalgia, and weakness.
- Outbreaks of illness of variable extent and severity occur nearly every year.
- Such outbreaks result in significant morbidity rates in the general population and in increased mortality rates among certain high-risk patients, mainly as a result of pulmonary complications.

Aetiology

1. Influenza is an acute systemic viral infection that primarily affects the respiratory tract and it carries a significant mortality.
2. It is caused by influenza A virus or influenza B virus (milder form).
3. Infection is seasonal, and variation in the haemagglutinin (H) and neuraminidase (N) glycoproteins on the surface of the virus leads to disease of variable intensity each year.
4. Nomenclature of influenza strains is based on these glycoproteins, e.g. H1 N1, H3N2 etc.
5. Genetic shift results in the circulation of a new influenza strain within a community to which few people are immune, potentially initiating an influenza epidemic or pandemic in which there is a high attack rate and there may be increased disease severity.

Clinical features

1. Incubation period -1-3 days
2. Uncomplicated disease leads to fever, malaise and cough.
3. Viral pneumonia may occur, although pulmonary complications are most often due to superinfection with *Strep.pneumoniae*, *Staph. aureus* or other bacteria.
4. Rare extra pulmonary manifestations include myositis, myocarditis, pericarditis and neurological complications (Reye's syndrome in children, encephalitis or transverse myelitis).

Diagnosis / Investigations

Acute infection is diagnosed by viral Culture, or by antigen or RNA detection (reverse transcription RT)-PCR) in a nasopharyngeal sample. The disease may also be diagnosed retrospectively on the basis of seroconversion.

Avian influenza

1. Avian influenza is caused by influenza A viruses with alternative haemagglutinin antigens, including the H5N1 strain.
2. These viruses have an increased ability to bind to lower respiratory tract epithelium, causing more severe disease with increased incidence of viral pneumonia and respiratory failure.
3. The majority of cases have occurred in individuals with a history of exposure to poultry, predominantly in South-east Asia.
4. Recent 'flu' seasons, cases have spread further west and infection has been identified in Europe in migrating birds and imported poultry.
5. Existing strains have been associated with infrequent person-to-person transmission but there is a concern that adaptation of an avian strain to allow effective person-to-person transmission is likely to lead to a global pandemic of life-threatening influenza.

Swine influenza

1. Occasional cases of influenza are transmitted from pigs to humans.
2. An outbreak of swine flu began in 2009,

initially in Mexico and then spreading around the world.

3. The causative strain was shown to be an H1N1 strain which showed significant genetic variation from human strains of H1N1.
4. Clinical features of infections with this strain are typical of influenza A infection, although some cases have more pronounced enteric features.

Infectious mononucleosis (IM) and Epstein-Barr virus (EBV)

- Epstein-Barr virus (EBV) is the cause of heterophile-positive infectious mononucleosis (IM), which is characterized by fever, sore throat, lymphadenopathy, and atypical lymphocytosis.
- EBV is also associated with several human tumors, including nasopharyngeal carcinoma, Burkitt's lymphoma, Hodgkin's disease, and (in patients with immunodeficiencies) B cell lymphoma. The virus is a member of the family Herpesviridae. The two types of EBV that are widely prevalent in nature are not distinguishable by conventional serologic tests.
- About 50% of infections result in typical IM.
- The virus is usually acquired from asymptomatic excretes via saliva, either by droplet infection or environmental contamination in childhood, or by kissing among adolescents and adults.
- EBV is not highly contagious..

Clinical features

1. IM has a prolonged and undetermined incubation period
2. Fever, headache and malaise.
3. This is followed by severe pharyngitis, which may include tonsillar exudates, and non-tender interior and posterior cervical lymphadenopathy.
4. Palatal petechiae, periorbital edema, splenomegaly, inguinal or axillary lymphadenopathy, and macular, petechial or erythema multiform rashes may occur.

5. In most cases fever resolves over 2 weeks, and fatigue and other abnormalities settle over a further few weeks.
6. Death is rare but can occur due to respiratory obstruction, haemorrhage from splenic rupture or thrombocytopenia, or encephalitis.

Diagnosis

- The white blood cell count is usually elevated and peaks at 10,000–20,000/L during the second or third week of illness.
- Lymphocytosis is usually demonstrable, with >10% atypical lymphocytes.
- The latter cells are enlarged lymphocytes that have abundant cytoplasm, vacuoles, and indentations of the cell membrane
- CD8+ cells predominate among the atypical lymphocytes.
- Low-grade neutropenia and thrombocytopenia are common during the first month of illness.

Liver function test

- Liver function is abnormal in >90% of cases.
- Serum levels of aminotransferases and alkaline phosphatase are usually mildly elevated.
- The serum concentration of bilirubin is elevated in 40% of cases.

Dengue

Etiology

1. The principal vector is the mosquito *Aedes aegypti*, which breeds in standing water, collections of water in containers, water-based air coolers and tyre dumps are a good environment for the vector in large cities.
2. *Aedes albopictus* is a vector in some South-east Asian countries.
3. There are four serotypes of dengue virus, all producing a similar clinical syndrome;
4. Homonymic immunity after infection with one of the serotypes is life-long, but heterotypic immunity against the other serotypes lasts only a few months after infection.

Clinical features of dengue fever

Prodrome : 2 days of malaise and headache

Acute onset : Fever, backache, arthralgias, headache, generalised pains ('break-bone fever'), pain on eye movement, lacrimation, anorexia, nausea, vomiting, relative bradycardia, prostration, depression, lymphadenopathy, scleral injection

Fever : Continuous or 'saddle-back', with break on 4th or 5th day and then recrudescence; usually lasts 7-8 days

Rash : Initial flushing faint macular rash in first 1-2 days. Maculopapular, scarlet morbilliform rash from days 3-5 on trunk, spreading centrifugally and sparing palms and soles, onset often with fever defervescence. May desquamate on resolution or give rise to petechiae on extensor surfaces.

Convalescence

- Slow

Complications

Minor bleeding from mucosal sites, hepatitis, cerebral hemorrhage or oedema, rhabdomyolysis.

1. The incubation period from being bitten by an infected mosquito is usually 2-7 days.
2. Asymptomatic infections are common but the disease is more severe in infants and the elder.
3. A morbilliform rash, which characteristically blanches under pressure, may occur, often as the fever is settling.
4. A more severe illness, called dengue haemorrhagic fever or dengue shock syndrome, occurs mainly in children in South-east Asia
5. In mild forms, there is thrombocytopenia and haemoconcentration.
6. In the most severe form, after 3-4 days of fever, hypotension and circulatory failure develop with pleural effusions, ascites, hypoalbuminaemia and features of acute respiratory distress syndrome.
7. Minor (petechiae, ecchymoses, epistaxis) or major (gastrointestinal or cerebrovascular) haemorrhagic signs may occur.

Diagnosis

1. Leucopenia is usual and thrombocytopenia common.

2. The diagnosis is confirmed by either at fourfold rise in the antibody titres, isolation of dengue virus from blood or detection of dengue virus RNA by PCR.
3. Serological tests may detect cross-reacting antibodies against other flavi viruses, including yellow fever vaccine.

Leptospirosis

- Leptospirosis is an infectious disease of humans and animals that is caused by pathogenic spirochetes of the genus *Leptospira*.
- It is considered the most common zoonosis in the world and is associated with settings of poor sanitation and agricultural occupations involving contact with animals or water.
- Increasingly, adventurous travel and "mud run" sports or races involving fresh water or soil exposure put humans at risk. Affected reservoir species experience chronic colonization of proximal renal tubules and shed leptospires in urine. Humans are not typically considered reservoirs but may chronically shed leptospires and may act as reservoirs in areas of high endemicity and poor waste sanitation.
- Leptospirosis is endemic in most areas where dengue virus is transmitted and may be mistaken for dengue, which is typically more common. Co-infection occurs in up to 8% of cases. Leptospirosis may also co-infect with dengue and other regionally prevalent pathogens, such as malaria, hantavirus, and scrub typhus.
- Leptospirosis is typically a biphasic pattern of early flu-like, septicemic illness followed by an inflammatory second phase. The latter may be characterized by systemic inflammatory response syndrome (SIRS) or cytokine storm. COVID-19 may follow a similar trajectory and may complicate diagnosis in the subtropics and tropics. These issues may complicate the diagnosis, especially in travelers or migrants presenting to non-endemic areas. It is important to consider leptospirosis when dengue is diagnosed in a severely ill patient, because early antibiotics are beneficial.

Etiology

- Leptospirosis is caused by spiral bacteria that belong to the genus *Leptospira*, the family Leptospiraceae, and the order Spirochaetales. These spirochetes are finely coiled, thin, motile, obligate, slow-growing aerobes.
- The traditional system divided the genus into 2 species: the pathogenic *Leptospira interrogans* and the nonpathogenic *Leptospira biflexa*. These species were divided further into serogroups, serovars, and strains based on shared antigens. *L. interrogans* included more than 250 serovars.
- Nevertheless, any leptospiral serovar can lead to the signs and symptoms seen with this disease. For example, jaundice occurs in 83% of patients with *L. icterohaemorrhagiae* infection and in 30% of patients infected with *L. pomona*. Aseptic meningitis commonly occurs in those infected with *L. pomona* or *L. canicola*.
- Transmission of leptospires to humans typically occurs by invasion across mucosal surfaces or nonintact skin. Infection may occur via direct contact with infected animals or their tissues or urine or through contact with contaminated water and soil.

Clinical presentations

- Onset of clinical illness occurs abruptly, after an incubation period of 2-30 days (typically 5-14 d).
- Expert consensus is that leptospirosis occurs as two recognizable clinical syndromes: anicteric and icteric (the existence of a third syndrome of asymptomatic infection is more controversial).
- Anicteric leptospirosis is a self-limited, mild flulike illness.
- Icteric leptospirosis, also known as Weil disease, is a severe illness characterized by multiorgan involvement or even failure.
- Two distinct phases of illness are observed in the mild form: the septicemic (acute) phase and the immune (delayed) phase. In icteric

leptospirosis, the 2 phases of illness are often continuous and indistinguishable. At disease onset, clinically predicting the severity of disease is not possible.

- An acute illness follows infection with any serovar of leptospirosis.

Most of the following signs and symptoms may develop in varying degrees:

- Headache
- Fever (38-40°C)
- Rigors
- Muscle pain (typically localized to the calf and lumbar areas)
- Nausea and vomiting
- Anorexia
- Diarrhea
- Dry cough
- Pharyngitis
- Conjunctival suffusion
- Nonpruritic skin rash

The natural course of leptospirosis falls into 2 distinct phases:

- The acute phase of illness lasts 5-7 days and is followed by a 1-3 day period of improvement in which the temperature curve falls and the patient may become afebrile and relatively asymptomatic. Subsequently, leptospirosis either regresses to a relatively asymptomatic illness or progresses to a more severe illness.
- Recurrence of fever indicates the onset of the second, immune stage. Nonspecific symptoms, such as fever and myalgia, may be less severe than in the first stage and last a few days to a few weeks. Many patients (77%) experience headache that is intense and poorly controlled by analgesics; this often heralds the onset of meningitis.

Investigations

- Leptospires grow slowly in culture, and recovery rates are low.
- Serologic tests are available only in specialized laboratories, and the sensitivity of acute serologic tests is low.

- Laboratory studies are used for two purposes: to confirm the diagnosis and to determine the extent of organ involvement and severity of complications. Laboratory confirmation of leptospirosis can be accomplished through isolation of the pathogen or by serologic testing.
- Isolation of the leptospires from human tissue or body fluids is the criterion standard, but culture is not routinely available.
- Urine is the most reliable body fluid to study because the urine contains leptospires from the onset of clinical symptoms until at least the third week of infection.
- Other body fluids contain the organism, but the window of opportunity to isolate them is shorter.
- Blood and CSF may produce positive PCR or cultures during the first 7-10 days of symptoms.
- Tissues (ie, liver, muscle, kidney, skin, eyes) are also sources of identification of the leptospires but are obviously more complicated to acquire.
- Antileptospire antibodies in these samples are detected using the microscopic agglutination test (MAT).

A complete blood cell count (CBC) is necessary. Findings on general laboratory studies are as follows:

- In patients with mild disease, elevated erythrocyte sedimentation rates and peripheral leukocytosis ($3,000-26,000 \times 10^9/L$) with a left shift are noted
- Significant anemia due to pulmonary and gastrointestinal hemorrhage can occur
- The platelet count may be diminished as a component of disseminated intravascular coagulation (DIC)
- Levels of blood urea nitrogen and serum creatinine may be profoundly elevated in the anuric or oliguric phase
- Serum creatine kinase levels (MM fraction) are often elevated in patients with muscular involvement.

- Coagulation times may be prolonged in patients with hepatic dysfunction and/or DIC. On liver function testing, serum bilirubin levels elevate as part of the obstructive disease due to capillaritis in the liver. Levels of hepatocellular transaminases are elevated less often and less significantly (usually $< 200 U/L$). Jaundice and bilirubinemia disproportional to hepatocellular damage is common in leptospirosis; alkaline phosphatase levels may be elevated 10-fold.
- On urinalysis, proteinuria may be present. Leukocytes, erythrocytes, hyaline casts, and granular casts may be present in the urinary sediment. Rapid urine antigen test using lateral flow assay is under development.
- Analysis of the CSF is useful only in excluding other causes of bacterial meningitis. When the CNS becomes involved in leptospirosis, polymorphonuclear leukocytes initially predominate and are later replaced by monocytes. CSF protein may be normal or elevated, whereas glucose levels remain normal. CSF pressure is normal, but a lumbar puncture can relieve the headache. Leptospires are routinely isolated from the CSF, but this finding does not change management of the disease.
- Imaging studies are useful in determining the extent and severity of organ involvement. This may include chest radiography or computed tomography to evaluate lung disease and biliary tract ultrasonography in suspected acalculous cholecystitis.
- Electrocardiographic (ECG) abnormalities are common during the leptospiremic phase of Weil syndrome. In severe cases, congestive heart failure and cardiogenic shock may occur.

Chikungunya virus

Aetiology

1. It is an alpha virus that causes fever, rash and arthropathy.
2. It is found principally in Africa and Asia, including India.

3. Humans and non-human primates are the main reservoir and the main vector is the *Aedes aegypti* mosquito.
4. Cases occur in epidemics on a background of sporadic cases.

Incubation period

The incubation period is 2-12 days.

Clinical features

1. A period of fever may be followed by an afebrile phase and then recrudescence of fever.
2. Children may develop maculo-papular rash.
3. Adults are susceptible to arthritis, which causes early morning pain and swelling, most often in the small joints.
4. Arthritis can persist for months and may become chronic in HLA-B27-positive individuals.
5. Related alpha virus causing similar syndromes include Sindbis virus (Scandinavia and Africa), O'nyong-nyong virus (Central Africa), Ross River virus (Australia) and Mayaro virus (Caribbean and South America).
6. A related virus, O'nyong-nyong, caused a major epidemic of arthritis and rash involving at least 2 million people as it moved across eastern and central Africa in the 1960s.
7. After its mysterious emergence, the virus virtually disappeared, leaving only occasional evidence of its persistence in Kenya until a transient resurgence of epidemic activity in 1997.

Diagnosis

- It is done by serology but cross-reactivity between alpha viruses occurs.
- The rash is most intense on the trunk and limbs and may desquamate.
- Petechiae are occasionally seen, and epistaxis is not uncommon, but this virus is not a regular cause of the HF syndrome, even in children.

Full blood count

- A few patients develop leukopenia.
- mildly decreased platelet counts

LFT

- Elevated levels of aspartate aminotransferase (AST) have been described

C-reactive protein

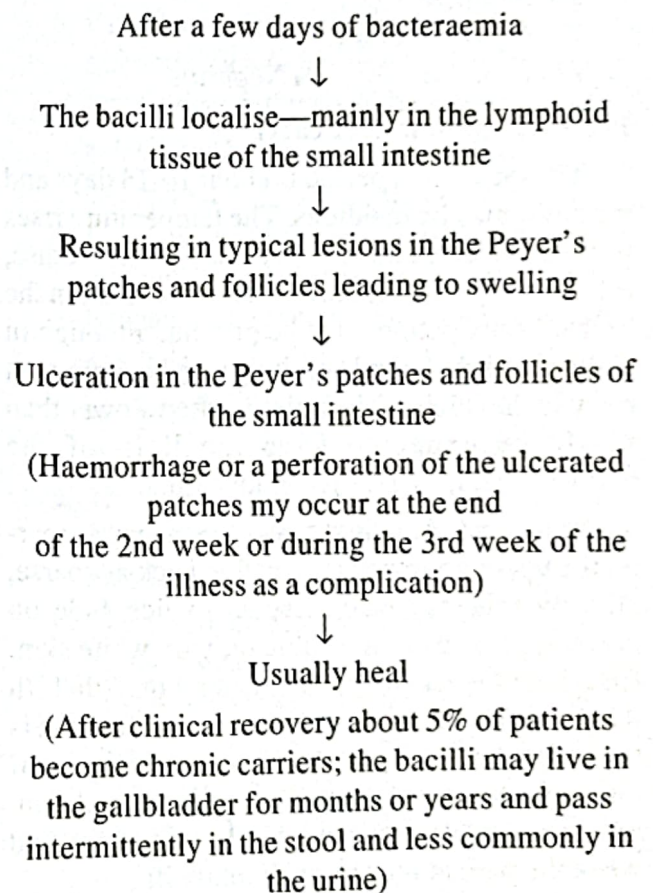
- Elevated levels of C-reactive protein have been described

Typhoid fever / Enteric fever

Etiology

Typhoid and paratyphoid fevers, which are transmitted by the faecal-oral route, are important causes of fever in India, sub-Saharan Africa and Latin America. Enteric fevers are caused by infection with *Salmonella typhi* and *S. Paratyphi A* and *B*.

Pathogenesis



Clinical features Typhoid fever

Clinical features of typhoid fever :

1st week

- Fever
- Relative bradycardia

- Headache
- Myalgia
- Constipation
- Diarrhoea and vomiting in children

End of 1st week

- Rose spots on turn
- Splenomegaly
- Cough
- Abdominal distension
- Diarrhoea

End of 2nd week

- Delirium, complications, then coma and death (if untreated)

Complications of typhoid fever

Bowel

- Perforation
- Hemorrhage

Septicaemic foci

- Bone and joint infection
- Cholecystitis
- Meningitis

Toxic phenomena

- Myocarditis
- Nephritis

Persistent gallbladder carriage

The incubation period is about 10-14 days and the onset may be insidious. The temperature rises in a step-ladder fashion for 4 or 5 days, with malaise, increasing headache, drowsiness and aching in the limbs. Constipation may be present, although in children diarrhoea and vomiting may be prominent early in the illness. The pulse is often slower than would be expected from the fever, of the temperature, i.e. a relative bradycardia.

At the end of the first week, a rash may appear on the upper abdomen and on the back as sparse, slightly raised, rose-red spots, which fade on pressure. It is usually visible only on white skin. Cough and epistaxis occur. Around the 7th-10th day the spleen becomes palpable. Constipation is then succeeded by diarrhoea and abdominal distension with tenderness. Bronchitis and delirium may develop. If untreated, by the end of the 2nd week the patient may be profoundly ill.

Paratyphoid fever

The course tends to be shorter and milder than that of typhoid fever and the onset is often more abrupt with acute enteritis. The rash may be more abundant intestinal complications less frequent.

Complications

1. Haemorrhage or a perforation of the ulcerated patches may occur at the end of the 2nd week or during the 3rd week of the illness
2. Additional complications may involve almost any viscous or system because of the septicaemia present during the 1st week. Bone and joint infection is common in children with sickle-cell disease.

Diagnosis/investigations

Culture (blood, bone marrow, intestinal secretions and stool)

- Other than a positive culture, no specific laboratory test is diagnostic for enteric fever.
- The definitive diagnosis of enteric fever requires the isolation of *S. typhi* or *S. paratyphi* from blood, bone marrow, other sterile sites, rose spots, stool, or intestinal secretions.
- The sensitivity of blood culture is only 40-80%.
- Bone marrow culture is 55-90% sensitive.
- Culture of intestinal secretions (best obtained by a noninvasive duodenal string test) can be positive despite a negative bone marrow culture.
- If blood, bone marrow, and intestinal secretions are all cultured, the yield is >90%.
- Stool cultures, while negative in 60-70% of cases during the first week, can become positive during the third week of infection in untreated patients.

Full Blood count

- In 15-25% of cases, leukopenia and neutropenia are detectable.
- Leukocytosis is more common among children, during the first 10 days of illness, and in cases complicated by intestinal perforation or secondary infection.

LFT and muscle enzyme levels

- Moderately elevated liver function tests and muscle enzyme levels.

Serologic tests

- The classic Widal tests for "febrile agglutinins", are available.
- None of these tests is sufficiently sensitive or specific to replace culture-based methods for the diagnosis of enteric fever in developed countries.

PCR and DNA probe assays

- PCR and DNA probe assays to detect *S. typhi* in blood have been identified but have not yet been developed for clinical use.

MCQs

1. Fever having twice daily spikes is
 - a. T.B
 - b. Leishmaniasis
 - c. Familial mediterranean fever
 - d. Amoebiasis
2. Saddle back fever is seen in
 - a. Salmonella infection
 - b. T.B
 - c. CMV infection
 - d. Dengue fever
3. Kissing's disease (Glandular fever) is caused by
 - a. Streptococcus
 - b. Staphylococcus
 - c. Salmonella
 - d. Epstein Barr Virus
4. Which one is correct in relation to malaria?
 - a. The swollen infected liver cell eventually bursts, discharging motile merozoites into the bloodstream to produce malaria
 - b. The swollen infected liver cell eventually bursts, discharging motile sporozoites into the bloodstream to produce malaria
 - c. A single sporozoite eventually may produce from 10,000 to >30,000 daughter merozoites.
 - d. Both A and C
5. Which statement is true?
 - a. The incubation period for *S. typhi* averages 1014 days but ranges from 321 days, depending on the inoculum size and the host's health and immune status
 - b. The incubation period for *S. typhi* averages 2024 days but ranges from 721 days, depending on the inoculum size and the host's health and immune status
 - c. The incubation period for *S. typhi* averages 21-28 days but ranges from 821 days, depending on the inoculum size and the host's health and immune status
 - d. The incubation period for *S. typhi* averages 3014 days but ranges from 921 days, depending on the inoculum size and the host's health and immune status
6. Patient with fever, relative bradycardia, and splenomegaly with rose spots, dry cough and lymphadenopathy could be diagnosed as
 - a. Chikungunya
 - b. Dengue
 - c. Typhoid
 - d. All the above
7. Radionuclide lymphoscintigraphic imaging of the limbs reliably demonstrates widespread lymphatic abnormalities in which diseases
 - a. Filariasis
 - b. Plague
 - c. Deep vein thrombosis
 - d. Syphilis
8. Which one is correct in relation to malaria?
 - a. The swollen infected liver cell eventually bursts, discharging motile merozoites into the bloodstream to produce malaria
 - b. The swollen infected liver cell eventually bursts, discharging motile sporozoites into the bloodstream to produce malaria
 - c. A single sporozoite eventually may produce from 10,000 to 30,000 daughter merozoites.
 - d. Both A and C

Answer

1.	b	2.	d	3.	d
4.	d	5.	a	6.	c
7.	a	8.	d		